

Power From 'Solar Grid'

There is an urgent need to modify policies under Jawaharlal Nehru National Solar Mission to speedily solve power problem in small towns and rural areas where the productive life of people ends with sunset. Here are some reasons and suggestions

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The most attractive and fascinating feature of solar energy is its omnipresence and availability for free. The current photovoltaic (PV) solar cells may work with just about 17 per cent to 18 per cent efficiency but what matters is their ability to transform solar energy into usable electrical power from a free non-polluting energy source.

Currently, PV solar is indeed the best option since no better technology breakthrough is in sight which could be used on a mass scale for the next few years. The credit therefore goes to the current government for taking a bold initiative and setting an ambitious target of 2500 megawatts of solar electric power by 2012 and 20,000 megawatts by 2020. The primary aim of Jawaharlal Nehru National Solar Mission (JNNSM) is to speedily augment inadequate electrical power hurting our inclusive development.

The sun delivers more energy to the earth in one hour than what the world currently uses in a year. Therefore it is our innovation in structuring policies that will enable us to realise our dream of extending productive life of our rural folks that today sadly ends with sunset.

Power shortage blues

The people worst affected by power shortage and its quality from the copper grid are those in smaller towns and rural regions, like farmers, artisans and their children. Working people suffer from productivity loss since rural economic activity currently stops at sunset and thereby each rural household loses close to 15 man-hours each day. Children also suffer since they can't



study at night or learn by watching television. Women suffer due to their inability to get informed from infotainment available non-stop via television.

Solar power can correct this since, unlike a copper grid, a solar grid is available even in deep hinterlands. Only right policies from the government will allow us to use the solar grid to achieve inclusive development rather than waiting for the copper grid that indeed is a long-term project with its associated operational delays.

However, our objective to improve power reliability and quality in small towns will not be realised unless the current policy undergoes important changes in order to quickly exploit and benefit from the omnipresent solar grid to augment the weakest links in the copper grid.

Policy lacunae

The current policy allows only large solar energy plants of over 1MW capacity to feed power into a 33kV grid. This portion of the grid is basically a low-impedance node and therefore has more stable power than the power available in small towns and fringe rural areas served from 3.3kV and a lower-voltage grid primarily since these have high line impedance. The feeder capacity is mostly inadequate

as one goes down to 440V feeders. Huge power losses in the grid are mostly at these fag-end delivery points. An impedance analysis of the power grid done by the IEEE reveals that injecting alternative power at high-impedance nodes improves the

resultant power quality than feeding solar power at low-impedance nodes.

That's why in nearly every country like Germany, Spain, US and China, grid feed is allowed even in 230V line from 5kW solar power generators. The biggest hardship and problem of all consumers in these areas is the black-out as well as poor-quality fluctuating power.

Big industrial customers drawing power from 33kV or higher grid point generally receive good-quality power. Those receiving power from low-voltage feeders today suffer the most. It is true that the solar energy delivered into the 33kV grid will provide additional power from solar plants but this addition of 1 or 2 per cent power into these low-impedance grid nodes will not unfortunately improve the quality of power that people in non-metro areas chronically suffer from.

Unlike India, every other country in the world allows solar power to be fed into the grid even at 230V. We need to understand why. The first important reason for this is to ensure adequate return on investment for every plant above 5 to 10kW capacity. This is possible only when the plant is fully loaded at all times due to the harvesting capacity of the plant throughout the time the sun is available.